

① **Find the no. of lead balls, each 1 cm in diameter that can be made from sphere of diameter of 12cm.**

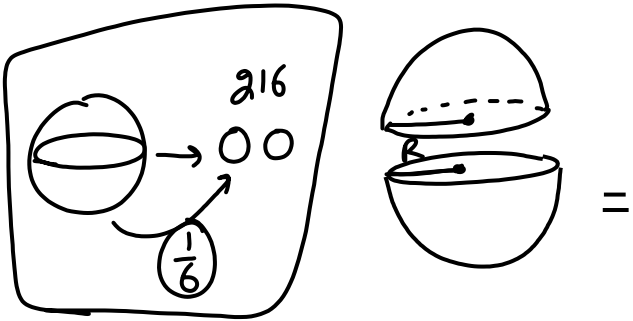
a) 144 **b) 1728**
c) 216 **d) 512**

(2) Let A and B be two solid spheres such that the surface area of B is 300% higher than the surface area of A. The volume of A is found to be k% lower than the volume of B. The value of k must be

B

?

a) 90.5 b) 87.5

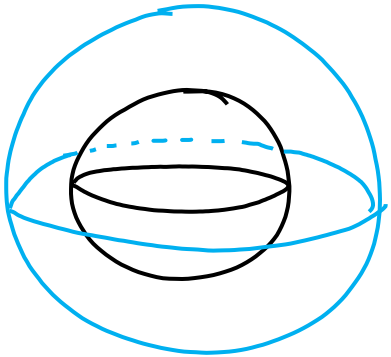


(3 V

(3) **If a hemisphere is melted and four spheres of equal volume are made, the radius of each sphere will be in comparison with hemisphere will be**

a) $\frac{1}{4}$ b) equal

c) $\frac{1}{2}$ d) $\frac{1}{6}$



(4) A ball of lead 4 cm in diameter is covered with gold. If the volume of the gold and lead are equal, then the thickness of gold [given $\sqrt[3]{2} = 1.259$] is approximately

$\sqrt{2} = 1.259$]

a) $15.038\ cm$ b) $0.259\ cm$

c) $1.036\ cm$ d) $0.518\ cm$

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(5) A large solid sphere is melted and moulded to form identical right circular cones with base radius and height same as the radius of the sphere. One of these cones is melted and moulded to form a smaller solid sphere. Then the ratio of the surface area of the smaller to the surface area of the larger sphere is

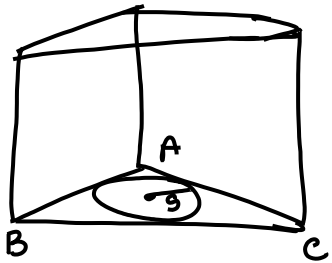
a) $1 : 3^{\frac{4}{3}}$

b) $1 : 2^{\frac{3}{2}}$

c) $1 : 3^{\frac{2}{3}}$

d) $1 : 2^{\frac{4}{3}}$

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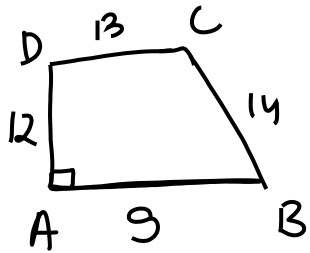


6 **The base of a right prism is a triangle of perimeter 45 cm and its incircle radius is 9 cm. Find its total surface area if its volume is 810 cm^3.**

a) 615 cm^2 b) 585 cm^2
c) 575 cm^2 d) 180 cm^2

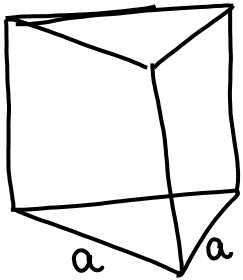
7 The base of a right prism is a quadrilateral ABCD such that $AB = 9$, $BC = 14$, $CD = 13$, $AD = 12$ and $\angle A = 90°$. Find the lateral surface area of the prism if its volume is 2070 cm³.

a) 672 b) 720
c) 768 d) 816

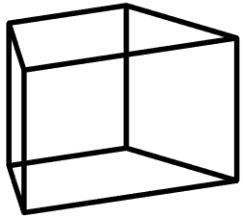


(9) The base of a right prism is an equilateral triangle. If the lateral surface area and volume is 120 $c\ m^2$, 40 $\sqrt{3}$ cm^3 respectively then the side of base of the prism is

a) $4\ cm$ b) $5\ cm$
c) $7\ cm$ d) $40\ cm$



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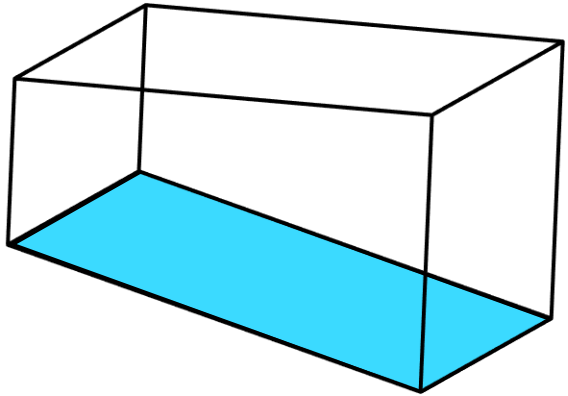


9) The base of a right prism is a trapezium, the length of whose parallel sides are $25\ cm$ and $11\ cm$ and the perpendicular distance between the parallel sides is $16\ cm$. If the height of the prism is $10\ cm$, then the volume of the prism is

?

a) 1440 cm b) 1540 cm

c) 2880 cm^3 d) 960 cm^3



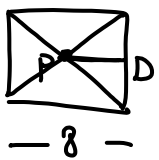
10 A swimming pool $9m$ wide and $12m$ long is 1m deep on the shallow side and $4m$ deep on the deeper side. Its volume is:

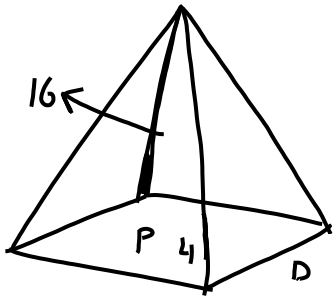
a)260 b)262
c)270 d)272

11 **A pyramid has a square base, whose side is 8cm. If the height of pyramids is 16 cm, then what is the total surface area of the pyramid?**

a) $64\left(\sqrt{17}+1\right)$ b) $32(\sqrt{13}+1)$

c) $64(2\sqrt{17}+1)$ d) $32(\sqrt{5}+1)$

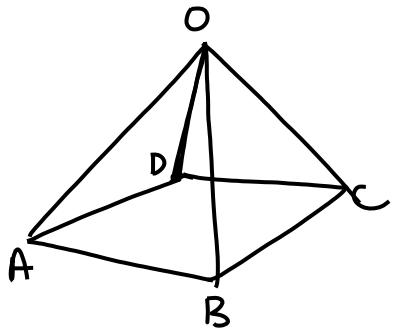




(12) The area of the square base of a right pyramid is $36\ cm^2$. If the area of each triangle forming the slant surface is $15\ cm^3$, then the volume of the pyramid is

a) $64\ cm^3$ b) $48\ cm^3$

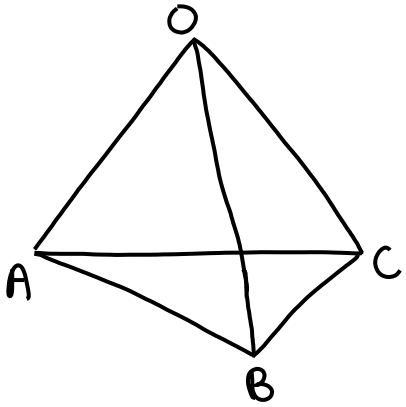
c) $24\ cm^3$ d) $144\ cm^3$



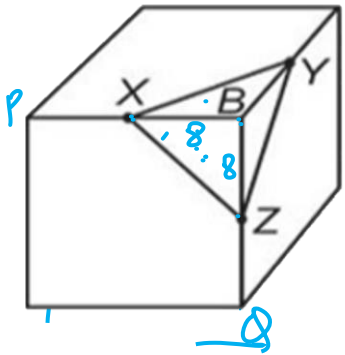
(13) The base of a right pyramid is an equilateral triangle with area $16\sqrt{3}cm^2$. If the area of one of its lateral faces is $30cm^2$, then its height (in cm) is:

a) $\sqrt{\frac{739}{12}}$ b) $\sqrt{\frac{209}{12}}$

c) $\sqrt{\frac{611}{12}}$ d) $\sqrt{\frac{643}{12}}$



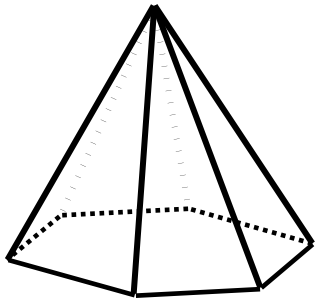
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14 A right triangular pyramid XYZB is cut from cube as shown in figure. The side of cube is 16 cm. X, Y and Z are mid points of the edges of the cube. What is the total surface area $(in\ cm^2)$ of the pyramid?

a) $48(\sqrt{3}+1)$ b) $24(4+\sqrt{3})$

c) $28(6+\sqrt{3})$ d) $32(3+\sqrt{3})$



15 **A regular hexagon base right pyramid has height $8cm$ and side of base is $4cm$. What is the total surface area of the pyramid?**

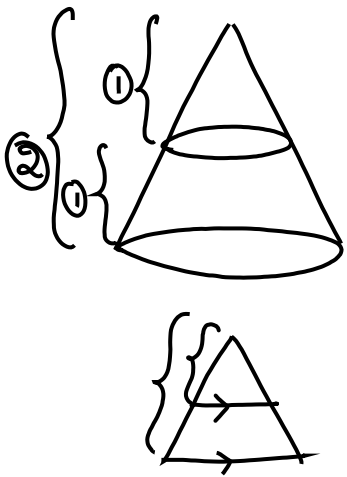
a) $24\left(\sqrt{13}+\sqrt{3}\right)$

b) $24\left(\sqrt{19}+\sqrt{3}\right)$

c) $24\left(2\sqrt{19}+\sqrt{3}\right)$

d) $48\left(\sqrt{19}+\sqrt{3}\right)$

)

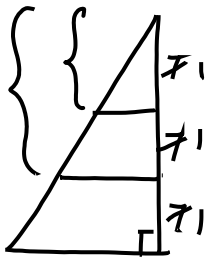


(16) **A plane divides a right circular cone into two parts of equal volume. If the plane is parallel to the base, then the ratio, in which the height of the cone is divided, is**

a) $1:\sqrt[3]{2}-1$ b) $1:\sqrt[3]{2}+1$

b) $1:\sqrt[3]{2}$ d) $1:\sqrt{2}$

(17) The radius of base of solid cone is 9cm and its height is 21 cm. It cut into 3 parts by two cuts, which are parallel to its base. The cuts are at height of 7 cm and 14 cm from the base respectively. What is the ratio of curved surface areas of top, middle and bottom parts respectively?

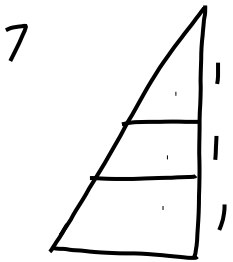


a) 1:4:8 b) 1:3:5
c) 1:3:9 d) 1:6:12

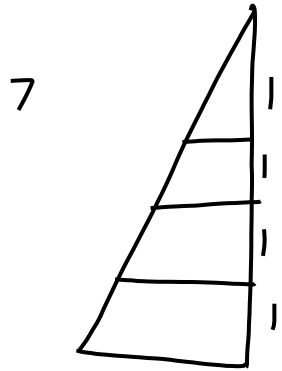
18

A regular triangular pyramid is cut by 2 planes which are parallel to its base. The planes trisects the altitude of the pyramid. Volume of top, middle and bottom part is $v_1 : v_2$ and v_3 respectively. What is the value of $v : v : v$?

a) 1:8:27 **b) 1:8:19**
c) 2:9:27 **d) 1:7:19**



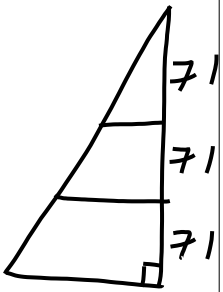
(19) **A right pyramid has a square base with side of base 12 cm and the height 40 cm. The pyramid is cut into four parts of equal heights by three cuts parallel to its base. What is the ratio of the volume of the four parts?**



a) 1:8:27:70 **b) 1:7:19:47**

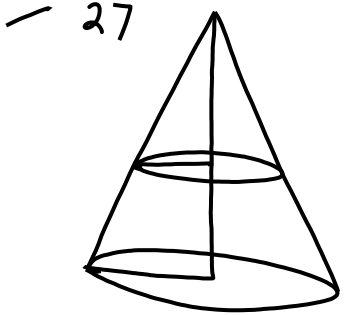
c) 1:7:19:37 **d) 1:8:27:64**

(20) A pyramid has square base. The side of square is 12 cm and height of pyramid is 21cm. The pyramid is cut into 3 parts by 2 cuts parallel to its base. The cuts are at height of 7 cm and 14 cm respectively from the base. What Is the difference between (in cm^3) in the volume of top most and bottom most part?



a) 672 b) 944
c) 786 d) 918

$= \frac{19}{27}$



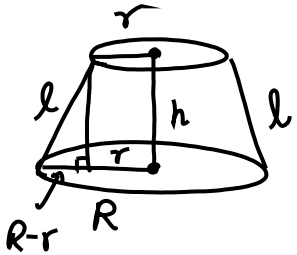
21 The height and radius of a right circular cone are respectively 21 cm and 9 cm. The cone is cut by a plane parallel to its base so as to divide it into two parts. The volume of the frustum of the cone is 1254 cm^3 . The radius of upper circular surface of frustum is

a) 4 cm b) 6 cm
c) 5 cm d) 7 cm

(21) The lateral surface area of frustum of a right circular cone, whose area of its base is $16\pi\ cm^2$, diameter of circular upper surface is $4\ cm$ and slant height is $6\ cm$, will be

a) $30\pi\ cm$ b) $48\pi\ cm$

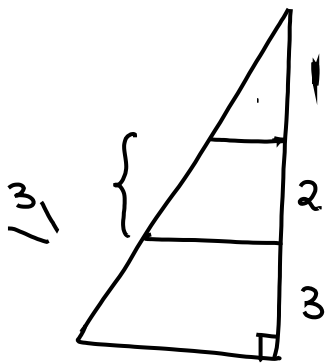
c) $36\pi\ cm^3$ d) $60\pi\ cm^3$



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(23) A cone of radius $90\,cm$ and height $120cm$ stands on its base. It is cut into 3 parts by 2 cuts parallel to its base such that the height of the three parts from top are in ratio of $1:2:3$. What is the total surface area of the middle part?

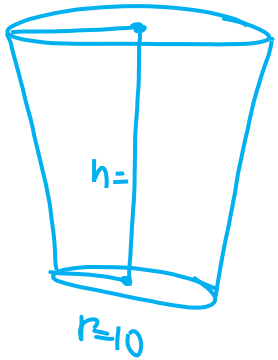
a)14600 b)16500
c)17800 d)18500



24 **A bucket is of a height 25 cm. its top and bottom radii are 20 cm and 10 cm, respectively. Its capacity (in L) is**

a) $\frac{17.5\pi}{3}$ b) 17.5π

c) 20π d) 25π



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